

1.

1.1 A self-driving car

Input: Camera images, GPS location, and sensor data.

Output: Steering, braking, and acceleration commands.

Justification: The car uses environmental data to determine how it should safely move.

1.2 Netflix recommendation system

Input: User's viewing history, ratings, and preferences.

Output: Recommended movies or TV shows.

Justification: Netflix uses user data to predict content the user is likely to enjoy.

1.3 Signature recognition

Input: An image or digital representation of a signature.

Output: Whether the signature is genuine or fraudulent.

Justification: The system compares the signature's features with known signatures.

1.4 Medical diagnosis

Input: Patient symptoms, medical history, and test results.

Output: A predicted disease or diagnosis.

Justification: The system uses patient information to identify the most likely medical condition.

2.

2.1 Classifying emails as promotional or social

Algorithm: Classification

Justification: The possible outputs are categories (promotional or social), rather than numerical values.

2.2 Forecasting stock price

Algorithm: Regression

Justification: Stock price is a continuous numerical value that can be predicted from historical data.

2.3 Sorting animal images into species

Algorithm: Classification

Justification: Each image is assigned to a specific category, such as dog, cat, or horse.

2.4 Predicting likelihood of a disease

Algorithm: Classification

Justification: The patient can be classified as likely to have the disease or unlikely to have it. If predicting an actual probability score, regression can also be used, but classification is the typical choice.

3.

3.1 Detecting manufacturing anomalies without known patterns

Algorithm: Unsupervised

Justification: There are no predefined labels for the anomalies, so the algorithm must identify unusual patterns on its own.

3.2 Predicting customer lifetime value

Algorithm: Supervised

Justification: Historical transaction data provides known customer values that can be used to train the model.

3.3 Segmenting customers based on behaviour

Algorithm: Unsupervised

Justification: The algorithm can discover groups of similar customers without predefined group labels.

3.4 Categorising social media posts into themes

Algorithm: Supervised

Justification: If posts have predefined theme labels, the algorithm can learn from those labelled examples and classify new posts.

4.

4.1 Fraudulent financial transactions — Most transactions labelled

Appropriate

Justification: Semi-supervised learning is useful when there is a combination of labelled and unlabelled data. The labelled transactions can guide the model while the unlabelled transactions provide additional information.

4.2 Customer satisfaction surveys — Only a small portion labelled

Appropriate

Justification: Semi-supervised learning is well suited because there are many unlabelled surveys and only a small number with satisfaction ratings.

4.3 Spam emails — Majority labelled

Inappropriate

Justification: Since most emails are already labelled, traditional supervised learning would generally be more appropriate. There is little benefit from semi-supervised learning when labelled data is abundant.

4.4 Credit card default prediction — Complete financial information

Inappropriate

Justification: If the dataset contains complete information and labelled examples of defaults/non-defaults, supervised learning would be more appropriate. Semi-supervised learning is mainly useful when labelled data is limited.